



UPI QuickPass Testing Guide for Acquirers
– Part I QuickPass online

V201905



May 2019

1 Summary of Change

Category	Change	Reason	Version
All	Updates testing card number from 7 to 17: 1. Remove QP6 2. Replace QP1 with QPC8, replace QP3 with QPC11, replace QP4 with QPC5, replace QP5 with QPC12, replace QP2 with QPC7, replace QP7 with QPC1 3. Add QPC2,QPC3,QPC4,QPC6,QPC9,QPC10,PC13,QPC14,QPC15,QPC16,QPC17	Simulate all the possible cards, make sure the POS terminals will have the correct behavior. Remove or replace redundant cards, add necessary cards	1.6
All	Separate functional test with integration test	Make the testing purpose clear	1.6
All	Expand functional test case	Expand functional test case due to the testing card number increases	1.6
All	Add test case related to pre -authorization in integration test	QuickPass shall support pre -authorization related transaction due to the development of mobile payment	1.6
All	Remove some test cases which use chip or MS in integration test	Some test cases in refund and cancelation use chip or MS way, remove them to simplify the test case.	1.6
All	F03 shall start from '20' in pre -authorization complete(request) cancelation message		1.6.1
All	Amend typo in this document		1.6.1
All	Add 81 BIN and modify wrong description	UPI will issue 81 BIN card	V201803
All	Change the signature base card expected result	Business rule changed	V201905

2 Effective date

New project which kicks off on and after September 30, 2019, shall use this test case.

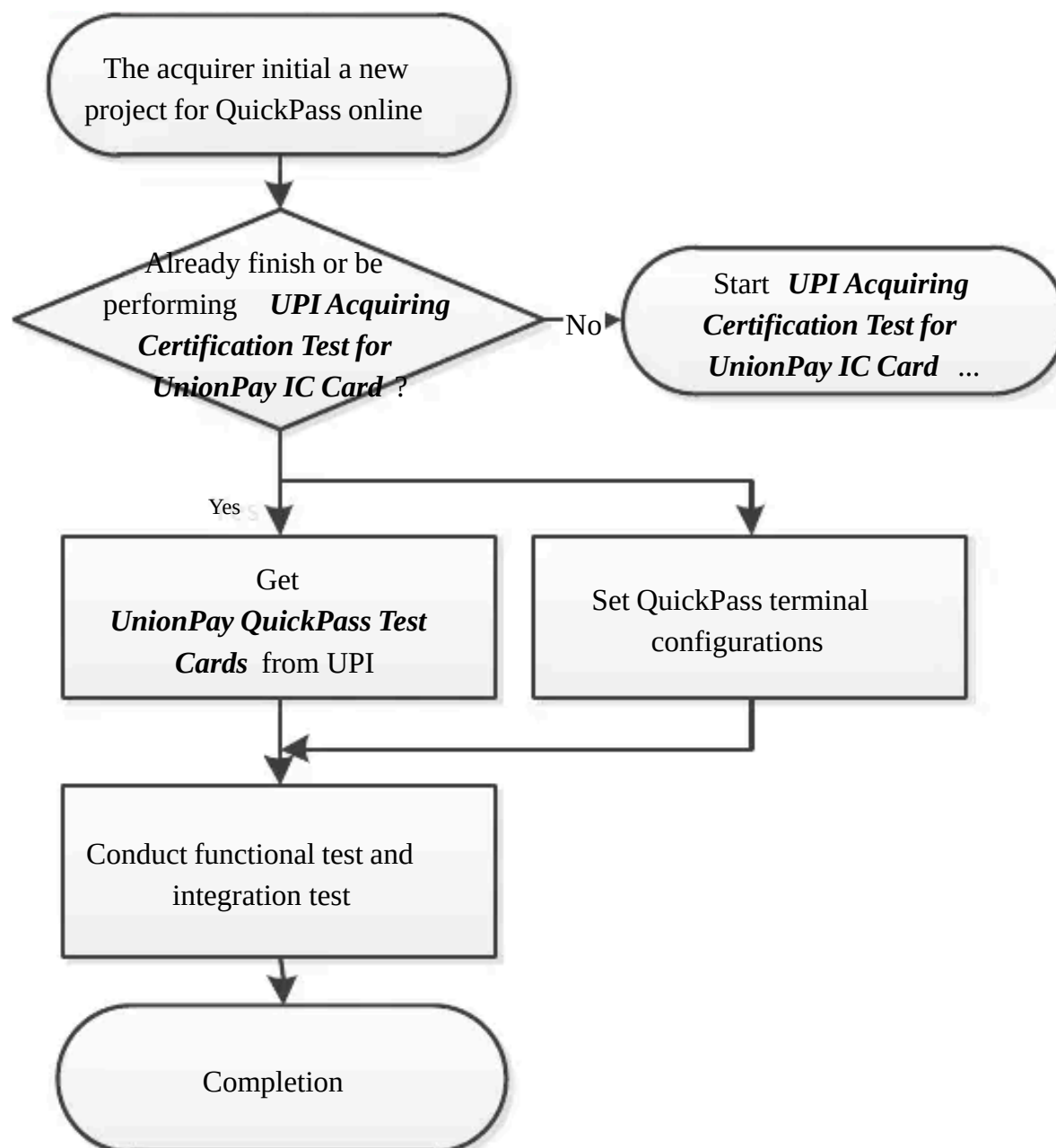
The old test cases version (V201803) will not be used after October 1, 2019.

3 Overview

This document aims to facilitate UnionPay International (hereinafter referred to as 'UPI') acquirers performing UPI Acquiring Certification Test for accepting UnionPay QuickPass cards by POS (Point of Sale) except Hongkong and Macao.

QuickPass online means all QuickPass transactions will authorized online. It allows issuer and acquirer to implement QuickPass online projects more convenient and quickly and also ensure the security of quick pay.

The following flowchart illustrates UPI Acquiring Certification Test procedure for an acquirer who initials a new QuickPass online acquiring project with UPI.



4 Precondition

Before the UPI Acquiring Certification Test for QuickPass online, acquirer shall complete UPI chip migration test, then start UnionPay QuickPass online Acquiring project, upgrade the acquiring system and obtain “ Terminal UICS Contactless Level 2 ” certificates for QuickPass online.

Acquirers shall finish level 3 additional modification:

Acquirers should request terminal vendors to perform addition modification. The modification is as below:

Terminals shall not ask for online PIN or signature when all of below situations are met

- QuickPass credit/Quasi credit cards are accepted,
- Contactless transaction amount is under the CVM limit of the terminal.

After modification, the terminal’s flow is as below:

1. Check AID(debit, credit, quasi -credit)
2. Perform the transaction processing:

AID	Acceptance process	
	<=Contactless QPS limit <CVM limit	>Contactless QPS limit >=CVM limit
Credit/ Quasi -Credit	No PIN No Signature	Follow the interaction result of card and terminal (EMV standard)
Debit	Follow the interaction result of card and terminal (EMV standard)	

5 Terminal Configurations

Besides the basic configurations for contact IC transaction, terminals (Point of Sale) need specific configurations for contactless transactions.

5.1 Terminal Transaction Qualifier (Tag ‘9F66’)

Terminal Transaction Qualifier shall be set as following:

Byte	Bit	Definition
1	8	0–RFU
	7	0–not supporting contactless UICC
	6	1–supporting contactless qUICC
	5	According to the type of terminals
	4	0–the terminal has the online capacity
	3	1–supporting the online PIN
	2	1–supporting the signature
	1	0-Reserved for future use

Byte	Bit	Definition
2	8	1-requiring the online cryptogram
	7	0 - do not request CVM by default
	6-1	0-Reserved for future use
3	8-1	Depending on terminals
4	8-1	0-Reserved for future use

5.2 Terminal Contactless Limit

Terminal Contactless Limit shall be set to 999999999999. The purpose is to make contactless transactions always available.

5.3 Terminal Floor Limit

Terminal Floor Limit shall be set to '0', all transactions will go online.

5.4 Terminal Currency Code

Acquirers shall prepare 2 terminals.

Terminal A: For currency matched test. Currency code shall be set as '156', RMB

Terminal B: For currency unmatched test. Currency code shall be set as local currency.

5.5 Reader CVM Required Limit

In production environment, Reader CVM Required Limit shall be set according to requirements of UPI OR (Annex 9, contactless payment, country -level transaction limit).

In functional, offline and online test, Reader CVM Required Limit shall be set to '300' for test convenience.

6 Test Cards

Card No.	Card Number (V1.1)	Card Number (V1.0)	Description
QPC1	6210945888010005	6210945888010005	Debit Card, QuickPass, Single Currency (RMB), PAN: 16 digits CVM – PIN + Signature(PIN preceding), Nonmatching Currency Transaction require CVM, CVM limit = 9999, Support Online ODA
QPC2	6210945888021	6210945888021	Debit Card, QuickPass, Single Currency (RMB), PAN: 13 digits CVM – PIN + Signature(PIN preceding), Nonmatching Currency Transaction not require CVM, CVM limit = 9999
QPC3	6210945888030003	6210945888030003	Debit Card, QuickPass, Single Currency (RMB) , PAN: 16 digits CVM – Signature, Nonmatching Currency Transaction not require CVM, CVM limit = 9999
QPC4	6210945888040000007	6210945888040000007	Debit Card, QuickPass, Single Currency (RMB) , PAN: 19 digits CVM – Signature, Nonmatching Currency Transaction require CVM, CVM limit = 9999
QPC5	6210946888053	6210946888053	Credit Card, QuickPass, Single Currency (RMB) , PAN: 13 digits CVM -PIN + Signature(PIN preceding), Nonmatching Currency Transaction require CVM, CVM limit = 9999
QPC6	6210946888060008	6210946888060008	Credit Card, QuickPass, Single Currency (RMB) , PAN: 16 digits CVM -PIN + Signature(PIN preceding), Nonmatching Currency Transaction not require CVM, CVM limit = 9999, 9F63 = 383031303030300080B00000000000 , Support Online ODA
QPC7	6210946888070000009	6210946888070000009	Credit Card, QuickPass, Single Currency (RMB) , PAN: 19 digits CVM -Signature, Nonmatching Currency Transaction not require CVM, CVM limit = 9999 9F63 = 383031303030300080B00000000000
QPC8	6210946888080006	6210946888080006	Credit Card, QuickPass, Single Currency (RMB) , PAN: 16 digits CVM -Signature, Nonmatching Currency Transaction require CVM, CVM limit = 9999

QPC9	6210946888090005	6210946888090005	Quasi Credit Card, QuickPass Single Currency (RMB) , PAN: 16 digits CVM -PIN + Signature(PIN preceding), Nonmatching Currency Transaction require CVM, CVM limit = 9999
QPC10	6210945888100004	6210945888100004	Common AID Card+Debit AID, QuickPass Single Currency (USD) , PAN: 16 digits CVM - PIN + Signature(PIN preceding), Nonmatching Currency Transaction not require CVM, CVM limit = 9999
QPC11	6210946888110001	6210946888110001	Credit Card, QuickPass, Dual Currency (HKD+RMB) , PAN: 16 digits CVM - Signature, Nonmatching Currency Transaction not require CVM, CVM limit = 9999, Support Online ODA
QPC12	6210945888120002	6210945888120002	Debit Card, QuickPass, Dual Currency (MOP+RMB) , PAN: 16 digits CVM -PIN + Signature(PIN preceding), Nonmatching Currency Transaction require CVM, CVM limit = 9999, Support Online ODA
QPC13	6210945888130001	6210945888130001	Debit Card, QuickPass, Single Currency (RMB) , PAN: 16 digits CVM - PIN + Signature(PIN preceding), Nonmatching Currency Transaction require CVM, CVM limit = 200
QPC14	6210946888140008	6210946888140008	Credit Card, QuickPass, Single Currency (RMB) , PAN: 16 digits CVM - PIN + Signature(PIN preceding), Nonmatching Currency Transaction not require CVM, CVM limit = 200
QPC15	6210946888150007	6210946888150007	Quasi Credit Card, QuickPass Single Currency (RMB) , PAN: 16 digits CVM - Signature, Nonmatching Currency Transaction require CVM, CVM limit = 9999
QPC16	6210946888160006	6210946888160006	Quasi Credit Card, QuickPass Single Currency (RMB) , PAN: 16 digits CVM -PIN + Signature(PIN preceding), Nonmatching Currency Transaction not require CVM, CVM limit = 9999, Support Online ODA
QPC17	6210946888170005	6210946888170005	Quasi Credit Card, QuickPass Single Currency (RMB) , PAN: 16 digits CVM - Signature, Nonmatching Currency Transaction not require CVM, CVM limit = 9999
QPC18	6210946888180004	6210946888180004	Credit Card, QuickPass, Single Currency (RMB) ,

			PAN: 16 digits CVM – CDCVM + Signature, Nonmatching Currency Transaction not require CVM, CVM limit = 9999, always CDCVM performed ☐ AID=A0000003330101021111111111111111
QPC19	6292600090881161	absent	Debit Card , QuickPass, Single Currency (RMB), PAN: 16 digits CVM - PIN, Nonmatching Currency Transaction not require CVM, CVM limit=9999
QPC20	81719999000000018	absent	Credit card, QuickPass, Single Currency (RMB), PAN: 16 digits CVM – PIN + signature, Nonmatching Currency Transaction not require CVM, CVM limit=9999
QPC21	8171999900000000021	absent	Quasi Credit card, QuickPass, Single Currency (RMB), PAN: 19 digits CVM - Signature, Nonmatching Currency Transaction not require CVM, CVM limit=9999
QPC22	81639999000010	absent	Debit Card, QuickPass, Single Currency (RMB), PAN: 13 digits CVM – PIN +signature, Nonmatching Currency Transaction not require CVM, CVM limit=9999

Note: If member has owned V1.0 Quickpass test card, only need borrow 81 BIN test cards (include QPC20, QPC21, QPC22) and QPC19 (used for test the PAR function) for Test.

7 Functional Test Cases for QuickPass

7.1 Currency matched test

Use **Terminal A** for currency matched test (**Terminal A: Currency code shall be set as ‘156’, RMB**).

Use **Terminal B** for currency matched test (**Terminal B : Currency code shall be set as local currency**).

Function test cases file: [_QuickPassAcquirersFunctionTestCases_V201905](#)

Terminal A		Terminal B	
Test cases sheet Name	Test Case	Test cases sheet Name	Test Case
TAIS_QPS	From TAIS_QPS_M_001 to TAIS_QPS_M_019	TAIS_QPS	From TAIS_QPS_U_001 to TAIS_QPS_U_028

8 Online Integration Test Cases for QuickPass online

Online test cases file: [JC Card Acquirer Integration Test Case For Online_201905](#)

Online Integration Test Cases for QuickPass Transaction					
Type	Online/ Offline	Message Type	Transaction Type	Transaction Items	QuickPass Acquirer Test cases sheet Name
QuickPass (QPS)	Online	Single	Purchase	Purchase	AIS_QPS_PER
				Purchase Cancellation	AIS_QPS_PVR
				Refund	AIS_QPS_CTH
			Pre-Authorization	Pre-authorization	AIS_QPS_PPA
				Pre-authorization Cancellation	AIS_QPS_PNP
				Pre-authorization Completion(Request)	AIS_QPS_ACR
				Pre-authorization Completion(Advice)	AIS_QPS_ACRA
				Pre-authorization Completion Cancelation (Request)	AIS_QPS_PCR
		Dual	Authorization (Confirmed Amount)	Authorization	AID_QPS_AUT
				Authorization Cancellation	AID_QPS_AUC
				Refund & Settlement	AID_QPS_SET
			Authorization (Estimated Amount)	Authorization	AID_QPS_AET
				Authorization Cancellation	AID_QPS_AEC
				Refund & Settlement	AID_QPS_SET
	Offline	N/A	Purchase ☐ QuickPass Offline ☐	Purchase	AIS_FQPS_PER
				Settlement	AIS_FQPS_SET

Note: QuickPass Offline only for HongKong and Macao, Member should apply 2 sets test card (include QP1- QP7 old version) for test

9 UAC Functional Test Report

1 Contactless Kernel Identification, Payment Application Information and Terminal Basic Data

Contactless kernel Level 2 Approval Reference Number

UnionPay Payment Application Name and Version

Terminal Type

Terminal Capability

Terminal Country Code

Terminal Application Version Number

Terminal Currency Code

Terminal Multiple Currency Conversion is supported?

2 UAC Functional Test Result<UPI QuickPass Testing Guide for Acquirers – Part I QuickPass online>

The following table is recorded the detail result on each test case

Passing Standard: Transaction success, PIN and receipt results are as expected.

Currency matched test uses table below:

Test No.	Test Result			Remark
	PASS	FAIL	N/A	
TAIS_QPS_M -001				
TAIS_QPS_M -002				
TAIS_QPS_M -003				
TAIS_QPS_M -004				
TAIS_QPS_M -005				
TAIS_QPS_M -006				
TAIS_QPS_M -007				
TAIS_QPS_M -008				

TAIS_QPS_M -009				
TAIS_QPS_M -010				
TAIS_QPS_M -011				
TAIS_QPS_M -012				
TAIS_QPS_M -013				
TAIS_QPS_M -014				
TAIS_QPS_M -015				
TAIS_QPS_M -016				
TAIS_QPS_M -017				
TAIS_QPS_M -018				
TAIS_QPS_M -019				
TAIS_QPS_M -020				

Currency unmatched test uses table below:

Test No.	Test Result			Remark
	PASS	FAIL	N/A	
TAIS_QPS_U -001				
TAIS_QPS_U -002				
TAIS_QPS_U -003				
TAIS_QPS_U -004				
TAIS_QPS_U -005				
TAIS_QPS_U -006				
TAIS_QPS_U -007				
TAIS_QPS_U -008				
TAIS_QPS_U -009				

TAIS_QPS_U -010				
TAIS_QPS_U -011				
TAIS_QPS_U -012				
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